I. INTRODUCTION

In the modern job market, companies receive hundreds to thousands of resumes for every job opening, making the initial screening process increasingly complex and time consuming. Recruiters often spend countless hours reading through resumes to shortlist candidates based on qualifications, experience, and skills. However, manual screening is prone to several limitations, including human error, inconsistency, unconscious bias, and fatigue, all of which can negatively impact the quality of candidate selection and lead to missed opportunities.

To overcome these limitations and improve the efficiency of the hiring process, organizations are turning toward automation and intelligent systems. One such innovative solution is the use of Machine Learning (ML) and Natural Language Processing (NLP) to automate resume screening. Machine learning models can be trained on large volumes of past hiring data to learn patterns, identify relevant information from resumes, and make intelligent decisions about candidate suitability. These models are capable of analyzing resume content, comparing it with job descriptions, and classifying resumes into categories such as shortlisted, not shortlisted, or needs review.

In this project, we propose a Resume Screening Model using Machine Learning that automates the evaluation of resumes and assists recruiters in identifying the most suitable candidates for a given job profile. The system begins with resume parsing, where resumes in PDF or DOCX format are converted into readable text. Then, using NLP techniques such as tokenization, stopword removal, lemmatization, and feature extraction methods like TF-IDF (Term Frequency-Inverse Document Frequency) or word embeddings, the textual content is transformed into a structured format suitable for model training.

The core of the model is a supervised learning algorithm, such as Logistic Regression, Random Forest, Support Vector Machine (SVM), or XGBoost, trained on labeled datasets of resumes that have been previously shortlisted or rejected. The model learns to differentiate between strong and weak resumes based on the features extracted. After training, it can accurately classify new resumes uploaded by users, thereby reducing the workload on HR personnel and ensuring a consistent, fair, and unbiased screening process.

Additionally, the system can provide insights such as the confidence level of classification, top matching keywords, and even a ranking system for multiple candidates. By integrating this model with an Applicant Tracking System (ATS) or a web-based interface, recruiters can streamline their operations and make data-driven hiring decisions in real time.

The primary goals of this project are to (1) reduce the manual effort required in resume screening, (2) improve the accuracy and fairness of candidate selection, and (3) demonstrate the power of machine learning in solving real-world HR challenges. This system also has the potential to be expanded in the future with advanced features such as semantic similarity analysis, chatbot integration for candidate interaction, or deep learning models that better understand the context and intent within resumes.

In summary, the proposed Resume Screening Model leverages machine learning and NLP to transform traditional hiring practices by automating resume analysis and classification. It presents a scalable and intelligent solution that not only saves time and effort but also enhances the overall effectiveness and fairness of the recruitment process. With ongoing advancements in AI and access to quality data, this project serves as a foundational step toward building smarter HR systems for the future.

II. LITERATURE REVIEW

The Resume Analyzer System is a machine learning-based tool designed to help job seekers enhance their resumes by aligning them with specific job requirements and industry standards. It features secure login and uses MongoDB to safely store sensitive data. By leveraging advanced NLP and ML techniques, the system analyzes resumes for language quality, structure, skill gaps, and keyword usage. It offers personalized suggestions to improve weak areas and adapt to evolving hiring trends through dynamic updates. Unlike static models, it provides real-time, job-specific feedback, supports diverse resume formats, and helps candidates increase their chances of getting noticed in applicant tracking systems.[1]

This study introduces an automated Resume Analyzer system aimed at reducing the manual effort involved in initial resume screening. Using NLP and text mining techniques, the system extracts important information such as name, contact details, experience, and qualifications from PDF resumes. It features a unique scoring mechanism that evaluates resumes based on key components like skills, interests, and declarations. The tool generates a resume score to aid both applicants and recruiters in assessing resume quality. Among various machine learning algorithms tested, the SVM model delivered the best performance, outperforming kNN, Naïve Bayes, Decision Tree, and Random Forest.[2]

This study investigates how Natural Language Processing (NLP) and Machine Learning can streamline the resume screening process by addressing common hiring challenges such as time constraints, bias, and high application volumes. Through extensive experimentation using various ML models, it was found that the Gaussian Naive Bayes algorithm, combined with the Bag-of-Words method, achieved an impressive accuracy of 99.5%. The research underscores the capability of NLP-based systems to efficiently filter resumes, assist recruiters in selecting top candidates, and enhance the overall hiring process by saving time and reducing bias.[3]

This study highlights the dual purpose of resume screening systems: identifying the best candidates and providing users with feedback on their resume quality. Traditional manual methods are often inefficient and prone to errors. To address this, the proposed system uses machine learning and natural language processing techniques to score and match resumes in real time. Techniques like TF-IDF and Cosine Similarity are employed to compare candidate data with job requirements. The system delivers instant results and is suitable for large-scale organizations, including administrative and government agencies, where high volumes of resumes are processed daily.[4]

This study focuses on automating the resume screening process using Natural Language Processing (NLP) and Machine Learning (ML) to evaluate candidates based on their qualifications, experience, and skills. Manual screening is replaced by real-time analysis involving tokenization, stemming, lemmatization, TF-IDF, and string matching for keyword extraction. The system classifies candidates into suitable job roles using ML models, with Support Vector Machine (SVM) achieving the highest accuracy among Logistic Regression, Random Forest, and k-Nearest Neighbors. The study emphasizes aligning candidate skills with appropriate job categories and aims to modernize recruitment through intelligent screening techniques.[5]

This study introduces a framework for online personality prediction to enhance the erecruitment process. By analyzing data from resumes, social media, and psychometric tests, the system uses machine learning and NLP techniques—such as BERT—for personality classification based on models like Big Five, Myers-Briggs, HEXACO, and Enneagram. Several machine learning algorithms, including logistic regression, SVM, Naive Bayes, kNN, Random Forest, XGBoost, and LSTM, are evaluated for accuracy. The proposed approach aims to help recruiters better match candidates to job roles, thereby improving hiring outcomes and organizational productivity.[6]

This study proposes a Bidirectional LSTM-based model for improving resume screening and classification in digital recruitment. The architecture effectively captures contextual and sequential information in resumes, and its performance is further enhanced through word embeddings. Tested on a diverse resume dataset, the model outperforms traditional methods in both accuracy and processing speed. The paper highlights the model's potential for real-world recruitment automation, emphasizing its scalability and adaptability across industries, making it a valuable tool for HR professionals and recruitment agencies aiming to streamline hiring processes.[7]

This research emphasizes the transformative role of machine learning in promoting diversity and inclusion within organizations by addressing biases in HR processes such as recruitment, performance evaluation, and talent management. It explores how ML algorithms can help reduce unconscious bias in job descriptions, resume screening, and candidate shortlisting. The study also highlights the use of data-driven insights to ensure fair employee assessments and identify high-potential talent based on merit. By proposing responsible ML implementation guidelines, the research advocates for ethical and inclusive technology use in HR, ultimately supporting more diverse, equitable workplaces.[8]

This study explores the use of advanced technologies like DistilBERT and ensemble learning for automated resume classification. DistilBERT, a cutting-edge NLP model, effectively extracts meaningful patterns from resume text, achieving a high classification accuracy of 94.16%. Ensemble learning, which combines predictions from multiple independently trained classifiers, attained an accuracy of 86.83%. Together, these methods enhance the efficiency and accuracy of resume screening, helping organizations manage large volumes of job applications and streamline their recruitment process for more effective talent acquisition.[9]

This study introduces a Resume Parser and Job Description Matcher system that utilizes NLP techniques—especially SpaCy—and machine learning algorithms like Support Vector Machines (SVM) to enhance candidate screening. The system parses and analyzes resumes and job descriptions to extract and match relevant skills, experiences, and qualifications. By incorporating keyword relevance and contextual understanding, the proposed model delivers improved matching accuracy over traditional methods, streamlining recruitment workflows and supporting more effective workforce management.[10]

This research proposes an automated resume screening solution using machine learning to address the inefficiencies and biases in traditional recruitment methods. Various ML models—including SVM, Naive Bayes, and XGBoost—are evaluated, with XGBoost demonstrating the best performance (AUC-ROC: 0.89). The system assigns scores to resumes based on job requirements, enabling ranked shortlisting. A user-friendly web application was developed to enhance accessibility and streamline the process. This approach aims to simplify recruitment, boost accuracy, and make hiring more efficient and objective.[11]

This project introduces an AI-driven recruitment system that leverages natural language processing (NLP) and machine learning to automate and optimize key stages of the hiring process. The system analyzes resumes, evaluates qualifications, and understands context through NLP for more accurate candidate assessments. It incorporates adaptive machine learning models that learn from past recruitment data to improve performance over time. Additionally, it automates interview scheduling based on availability, enhancing efficiency. Designed in collaboration with HR professionals and data scientists, the system offers a smart, ethical, and data-driven approach to modern talent acquisition.[12]

This study presents a novel approach to resume classification using Heterogeneous Graph Neural Networks (HGNN), where resumes, jobs, and job categories are represented as interconnected nodes in a heterogeneous graph. By treating resumes as target nodes and employing graph neural networks for feature aggregation and classification, the model effectively captures complex relationships among different entities. Experimental results demonstrate that this method outperforms traditional classification techniques, offering a promising direction for building more accurate and intelligent resume screening systems.[13]

Recruitment processes have evolved from manual sorting of the resumes to automation with the integration of Natural Language Processing. The benefits of NLP in hiring new employees are examined in this paper, with a focus on how it might increase candidate selection's effectiveness, objectivity, and fairness. This automated approach streamlines the candidates resume evaluation process. The core steps in the system include parsing resume to text, performing NLP of parsed text, creating a NER (Named Entity Recognition) model and using NER model to calculate P, R and F score to generate final score to sort.[14]

This paper introduces a two-in-one interview application that automates the Behavioral Event Interview (BEI) process using Natural Language Processing (NLP) and Machine Learning. The system reduces interviewer workload and company expenses by scanning resumes, eliminating biased information like caste and gender, and generating both behavioral and technical interview questions. It uniquely focuses on assessing both candidate behavior and resume qualifications, making it a pioneering tool in early-stage recruitment screening to filter out unsuitable candidates efficiently.[15]

This study addresses the inefficiencies and biases in manual resume screening by leveraging Natural Language Processing (NLP) and machine learning for automated classification and matching. Resumes, often submitted in diverse formats and styles, are preprocessed and vectorized using TF-IDF. The classification process evaluates the performance of three machine learning models—Bidirectional LSTM (Bi-LSTM), Multinomial Naive Bayes (MNB), and K-Nearest Neighbors (KNN)—on a large, labeled resume dataset. To enhance the matching process, cosine similarity scores are calculated between resume and job description vectors. This two-step approach improves classification accuracy and ensures a more efficient and objective alignment of candidates with job roles.[16]

Resumes, submitted in various formats and styles, pose challenges for manual screening due to their inconsistency and volume. Traditional methods are time-consuming, biased, and often inefficient. To address this, the study explores the use of NLP and machine learning algorithms—Bi-LSTM, Multinomial Naive Bayes, and K-Nearest Neighbors—for resume classification. Resumes are vectorized using TF-IDF, and cosine similarity is used to match them with job descriptions. This two-step approach improves classification accuracy and streamlines the recruitment process.[17]

Traditional resume screening is still used by some HR teams but is time-consuming and manual. While larger firms use Applicant Tracking Systems (ATS), newer methods like semantic search offer deeper contextual understanding beyond keyword matching. With 50–95% accuracy, semantic search improves resume screening. Integrating Large Language Models (LLMs) using the LangChain framework boosts system scalability and eliminates the need for retraining, enhancing efficiency and flexibility in the hiring process.[18]

This paper explores automated resume parsing and evaluation using NLP and machine learning. Traditional manual methods are slow and error-prone, while modern techniques like Named Entity Recognition (NER) and cosine similarity enable efficient screening. We review existing frameworks, implement our own models on third-party resume datasets, and assess performance using statistical analysis and visualizations. The study offers insights valuable to researchers and recruiters in optimizing resume screening processes.[19]

This research proposes an efficient resume sorting system using Machine Learning and NLP, tailored to company-specific job requirements. The model uses cosine similarity to extract keywords from resumes and match them with recruiter-defined criteria. Trained on a diverse resume dataset, the system filters candidates by relevance, streamlining the hiring process and improving selection accuracy.[20]

III. PROBLEM STATEMENT

With the rapid growth of technology and digital transformation, job applications have increased exponentially, making it increasingly difficult for organizations to efficiently identify the most qualified candidates. Traditional resume screening methods involve manual efforts that are not only time-consuming but also prone to human errors and biases. Even though some companies use basic automated tools like Applicant Tracking Systems (ATS), these often rely on rigid keyword matching, which lacks contextual understanding and may lead to the rejection of potentially suitable candidates.

Moreover, resumes are submitted in various formats (PDF, DOC, TXT) and styles, making uniform evaluation a complex task. Many existing systems fail to adapt to dynamic job requirements or provide a comprehensive analysis of resume-job fit. As a result, recruiters often face challenges in filtering out unqualified candidates, ranking applicants appropriately, and ensuring fairness in the hiring process.

Therefore, there is a pressing need for an intelligent, scalable, and context-aware resume screening solution that leverages Machine Learning (ML) and Natural Language Processing (NLP) techniques. Such a system must be capable of parsing resumes, understanding the semantic meaning of job descriptions, extracting relevant information, and accurately matching candidate profiles to employer needs. Addressing this challenge will not only streamline the hiring process but also improve recruitment outcomes and reduce organizational workload.

IV. SYSTEM REQUIREMENT

▪ SOFTWARE REQUIREMENTS

- **Operating System**: Windows 10 / Linux / macOS
- **Programming Language**: Python (version 3.8 or above)
- **Web Framework**: Django
- **Frontend Technologies**: HTML, CSS, JavaScript
- **Database**: MySQL
- **Code Editor**: Visual Studio Code or PyCharm
- **Browser**: Google Chrome or Mozilla Firefox

Python Libraries Used:

- pymysql – for database connectivity with MySQL
- pandas – for data handling and manipulation
- PyPDF2 – to read and extract text from PDF resumes
- textract – for extracting text from various document formats
- matplotlib – to generate pie charts for visual analysis
- PIL (Pillow) – for basic image handling

- re, string, io, warnings, datetime – standard Python modules for data processing

HARDWARE REQUIREMENTS

- **Processor**: Minimum Intel i3 or equivalent
- **RAM**: At least 4 GB (8 GB recommended for smooth performance)
- **Storage**: 100 GB free space (SSD recommended)
- **Display**: Any screen with at least 1366x768 resolution
- **Internet**: Required for downloading libraries and tools

V. SYSTEM DESIGN

A. USE-CASE DIAGRAM

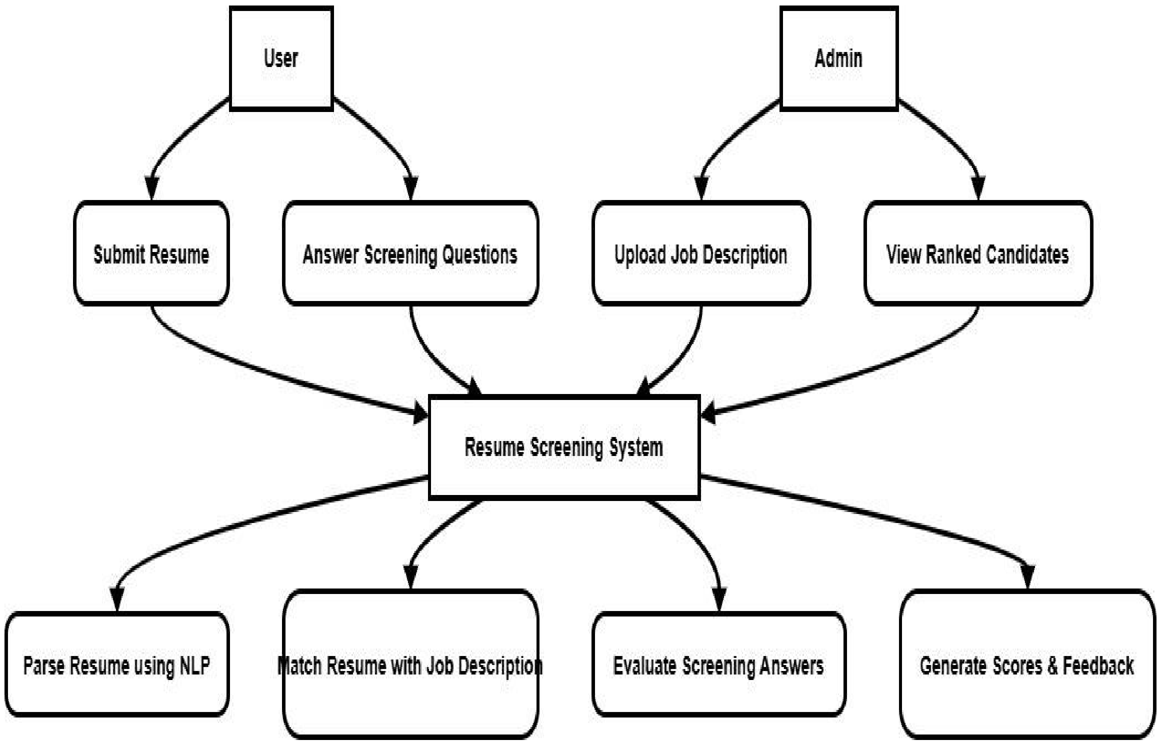


Fig 1. Use case diagram

- User can:
  - Submit their Resume.
  - Answer Screening Questions.

Admin can:

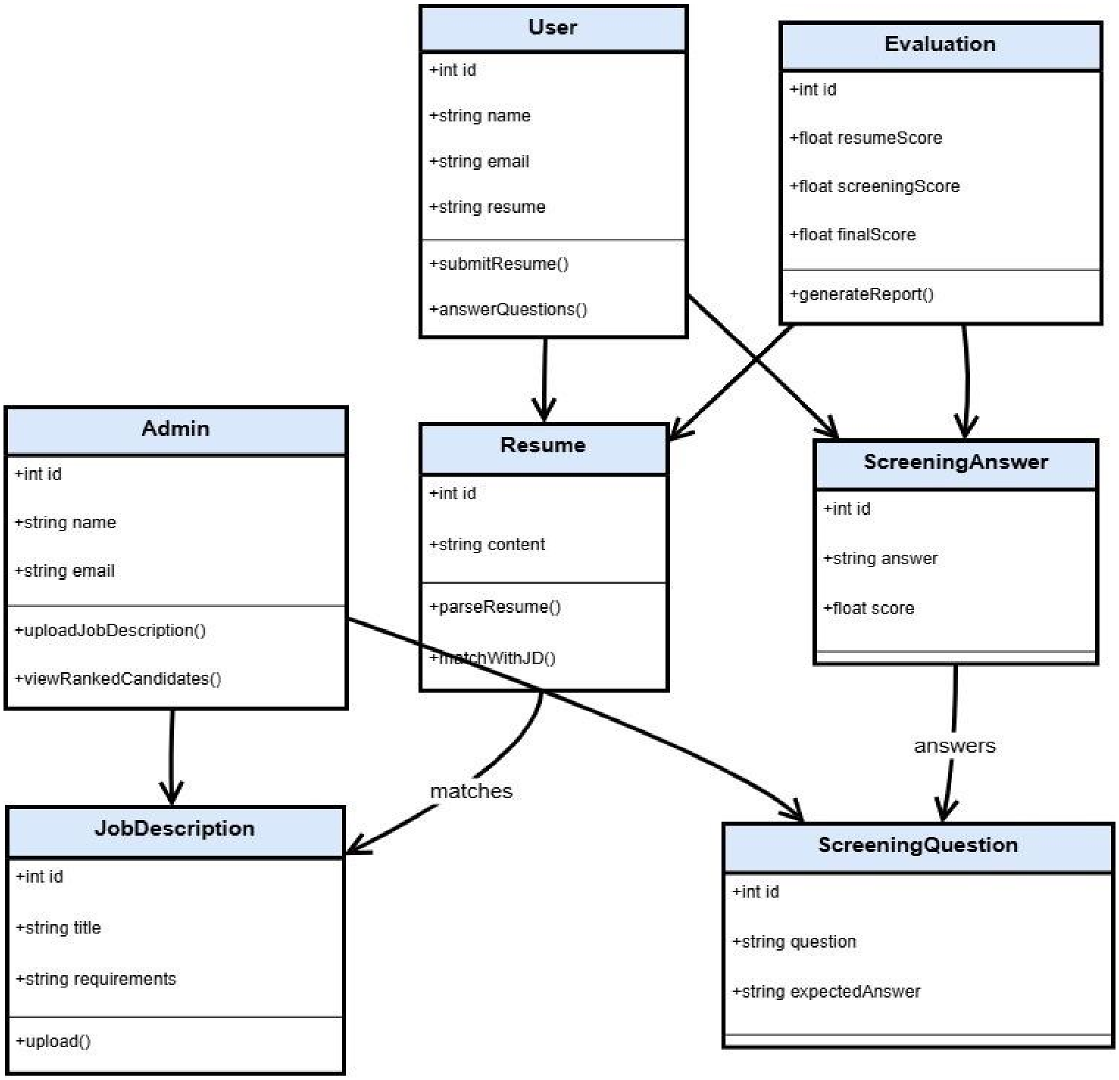
  - Upload the Job Description (JD).

- View the list of Ranked Candidates.

All these actions are handled by the Resume Screening System, which:

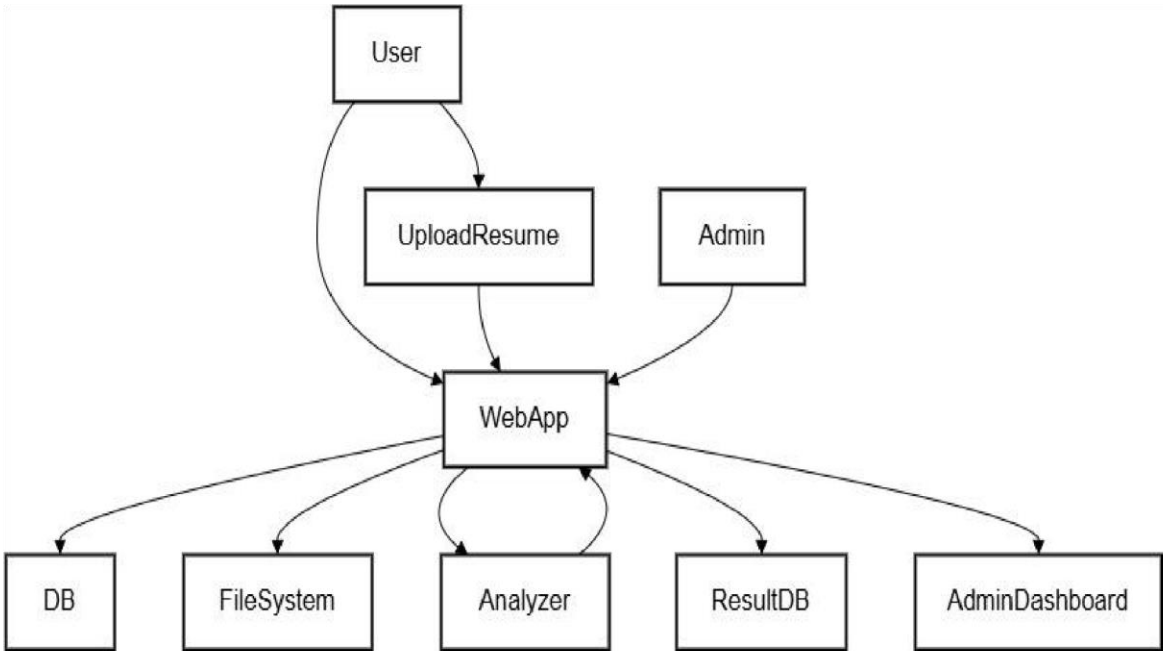
- Parses the resume using NLP.
- Matches resumes with job descriptions using ML/NLP techniques.
- Evaluates the answers to screening questions.
- Generates final scores and feedback.

B. CLASS DIAGRAM



- User submits Resume and answers Screening Question.
- Admin uploads Job Description and prepares screening questions.
- Resume is parsed and matched with Job Description.

C. DATA-FLOW DIAGRAM



- User → End user who interacts with the system
- WebApp → Django application logic
- DB → User registration/login data (MySQL)
- UploadResume → User uploads PDF resume
- FileSystem → Stores uploaded resume files
- Analyzer → Python-based resume keyword analysis logic
- ResultDB → Stores keyword match scores/results
- AdminDashboard → Admin's resume analytics view

D. SEQUENCE DIAGRAM

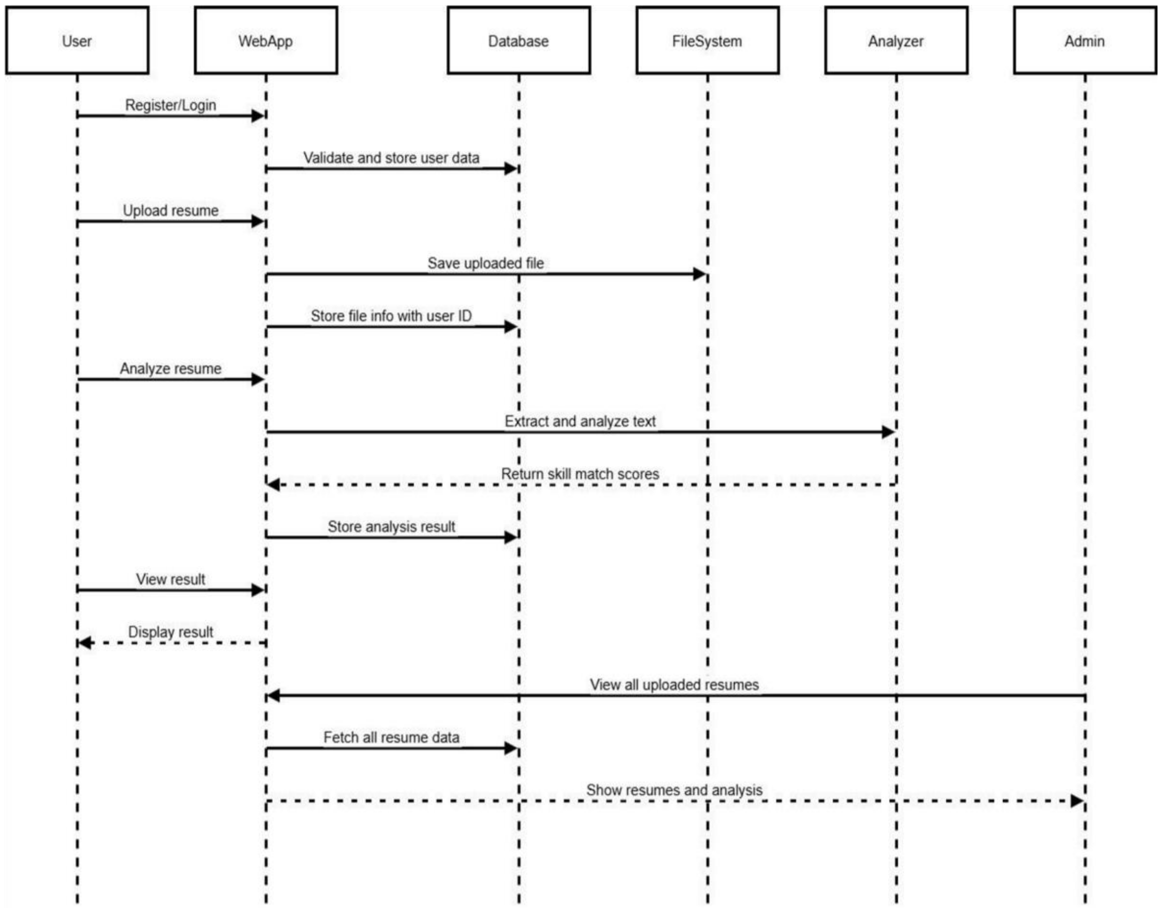


Fig 4.Sequence Diagram

- The flow starts with the user registering or logging in.
- They upload a resume, which gets saved and analyzed.
- Results are stored and displayed.
- Admin has access to all resumes and their analytics.

VI. WORKING OF SYSTEM

This system is developed using Django and MySQL, designed to automate the resume screening process and help recruiters match suitable candidates with relevant job profiles.

1. User Registration and Login

- Users can **register** using a form (/register view) where their name, contact number, email, and password are stored in the database.
- After registration, they can **login** (/dologin view) using their credentials.
- If the admin logs in with admin/admin, they are redirected to the **admin dashboard**.

2. Resume Upload Module

- Authenticated users can upload their resume as a **PDF file** through the /uploadImage view.
- The uploaded file is saved in the static/ folder, and the filename along with user details is inserted into the upload table in MySQL.

3. Resume Extraction and Analysis

- The system extracts **text from the uploaded PDF** using PyPDF2 (/analyze view).
- It removes numbers, punctuation, and converts everything to lowercase.

- It compares the extracted text with predefined **keywords** related to different domains: ○ Quality/Six Sigma ○ Operations Management ○ Supply Chain ○ Project Management
    - Data Analytics ○
      Healthcare

- It counts the number of keyword matches for each domain to evaluate the candidate's expertise area.

- A **pie chart** is generated using matplotlib to visualize the domain-wise decomposition of the resume.

4. Admin Dashboard

- The admin can view:
    - All users and uploaded resumes(/viewuser)
    - Analyze resumes(/resumeanalytic)
    - Select suitable candidates(/selectcandidate)
    - Manage uploaded resume(/manage)
- Admin can also open and view any resume PDF using the /pdf view.

5. Candidate Selection

- Admin can **filter and shortlist** candidates based on their domain-wise scores.

- Selected candidates are listed under the /selectedlist view.
- Not selected candidates or fallback handling goes to /notselected.

6. Database Structure

- **user table** stores user registration details.
- **upload table** stores uploaded resume file paths along with user information.
- The backend uses pymysql to handle all database operations.

Flowchart:

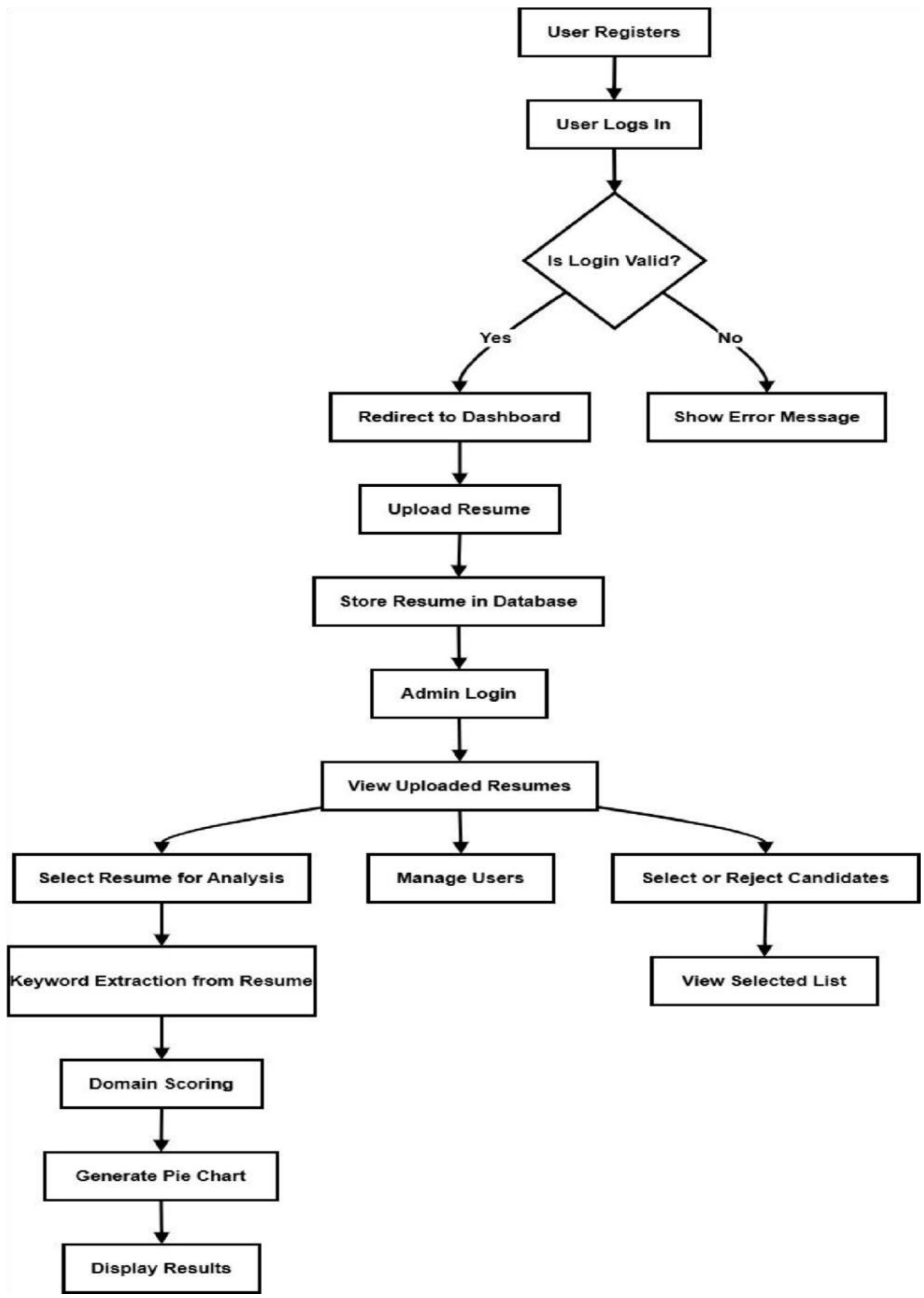


Fig 5. Flowchart Diagram

VII. PROJECT LAYOUT/ RESULT

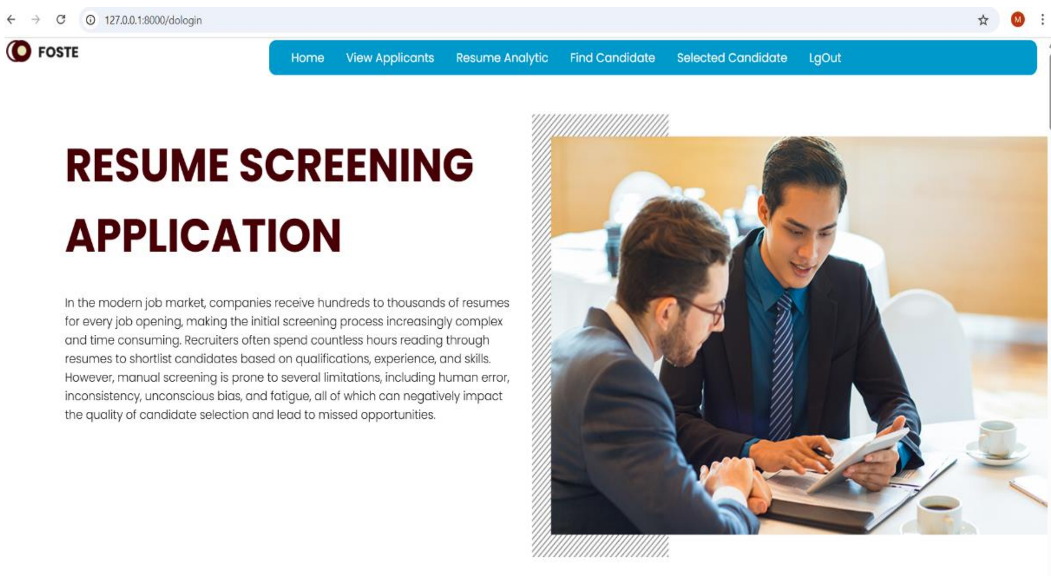


Fig 6. Home page of application

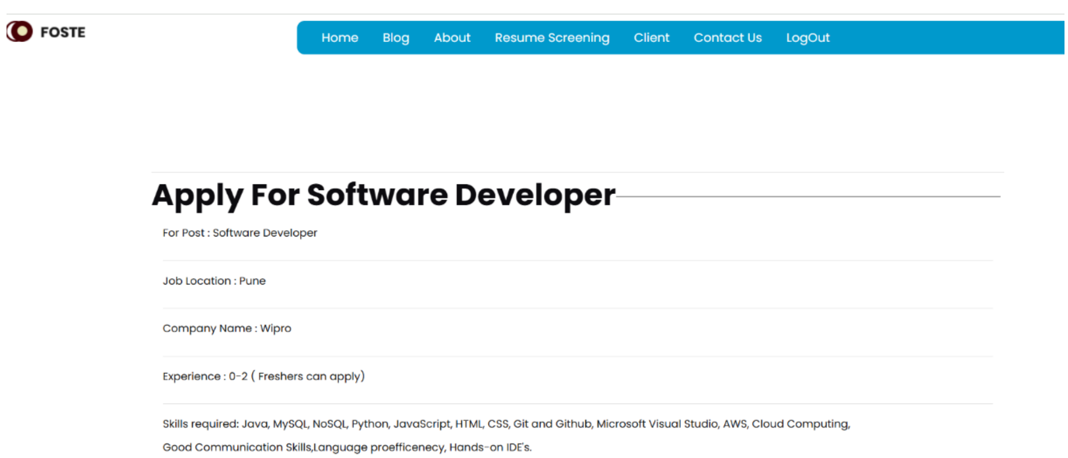


Fig 7. User interface

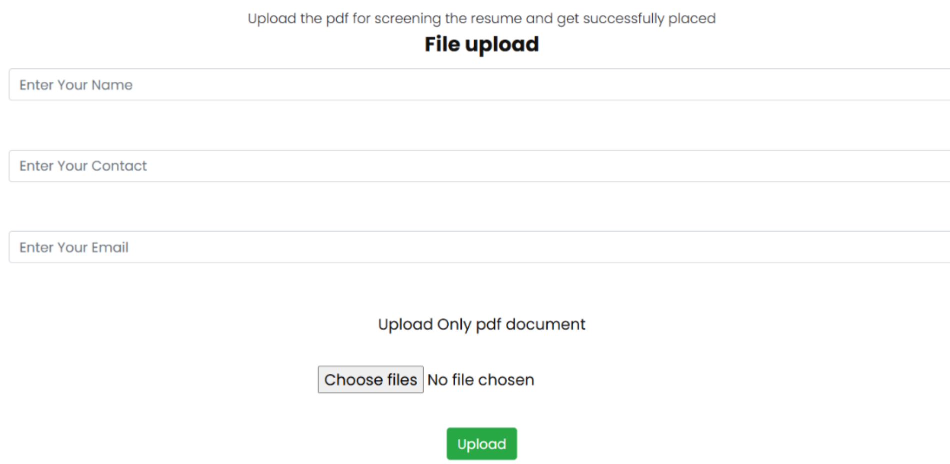
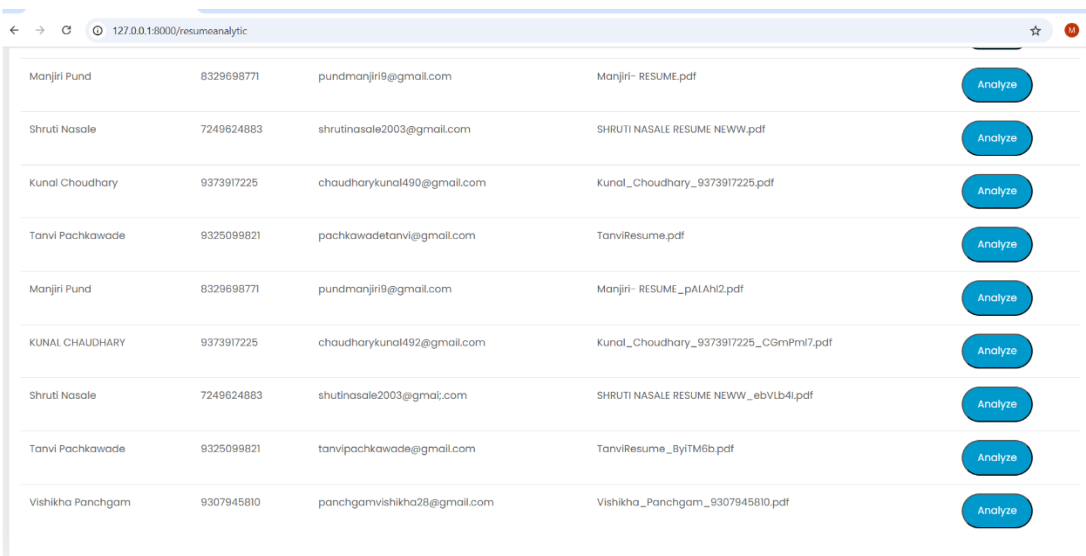
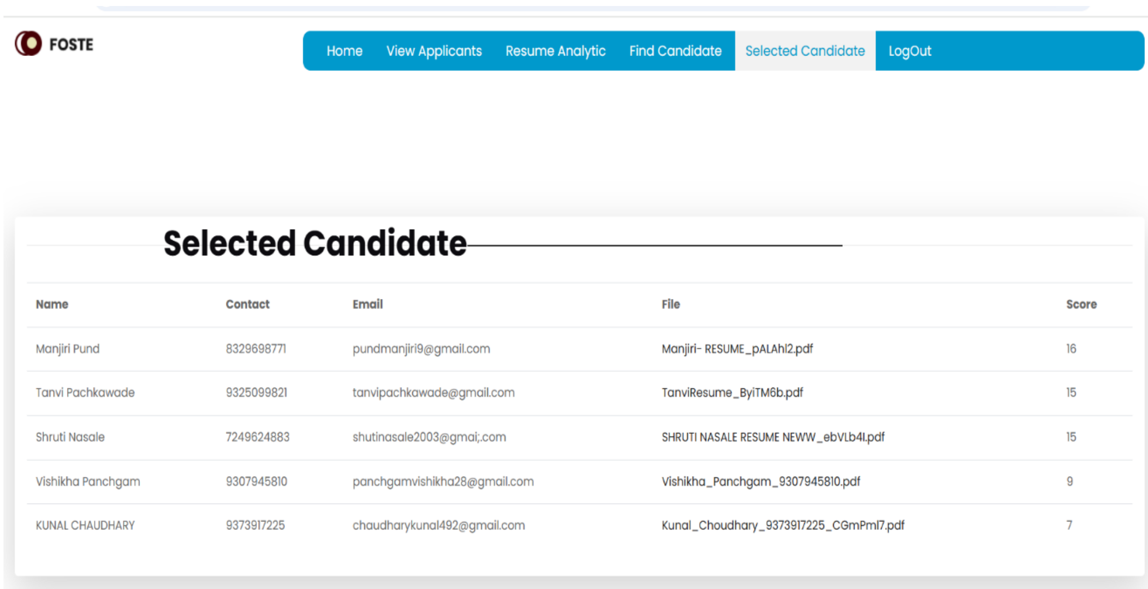


Fig 8.Uploading of Resume (user-Interface)



Manjiri Pund	8329698771	pundmanjiri9@gmail.com	Manjiri- RESUME.pdf	Analyze
Shruti Nasale	7249624883	shrutinasale2003@gmail.com	SHRUTI NASALE RESUME NEWW.pdf	Analyze
Kunal Choudhary	9373917225	chaudharykunal490@gmail.com	Kunal_Choudhary_9373917225.pdf	Analyze
Tanvi Pachkawade	9325099821	pachkawadetanvi@gmail.com	TanviResume.pdf	Analyze
Manjiri Pund	8329698771	pundmanjiri9@gmail.com	Manjiri- RESUME_pALAhl2.pdf	Analyze
KUNAL CHAUDHARY	9373917225	chaudharykunal492@gmail.com	Kunal_Choudhary_9373917225_CGmPml7.pdf	Analyze
Shruti Nasale	7249624883	shutinasale2003@gmai;.com	SHRUTI NASALE RESUME NEWW_ebVLb4l.pdf	Analyze
Tanvi Pachkawade	9325099821	tanvipachkawade@gmail.com	TanviResume_ByiTM6b.pdf	Analyze
Vishikha Panchgam	9307945810	panchgamvishikha28@gmail.com	Vishikha_Panchgam_9307945810.pdf	Analyze

Fig 9. Admin page(List of Uploaded Candidates)



Selected Candidate

Name	Contact	Email	File	Score
Manjiri Pund	8329698771	pundmanjiri9@gmail.com	Manjiri- RESUME_pALAhl2.pdf	16
Tanvi Pachkawade	9325099821	tanvipachkawade@gmail.com	TanviResume_ByiTM6b.pdf	15
Shruti Nasale	7249624883	shutinasale2003@gmai;.com	SHRUTI NASALE RESUME NEWW_ebVLb4l.pdf	15
Vishikha Panchgam	9307945810	panchgamvishikha28@gmail.com	Vishikha_Panchgam_9307945810.pdf	9
KUNAL CHAUDHARY	9373917225	chaudharykunal492@gmail.com	Kunal_Choudhary_9373917225_CGmPml7.pdf	7

Fig 10. Selected Candidates with an maximum no of scorer at top

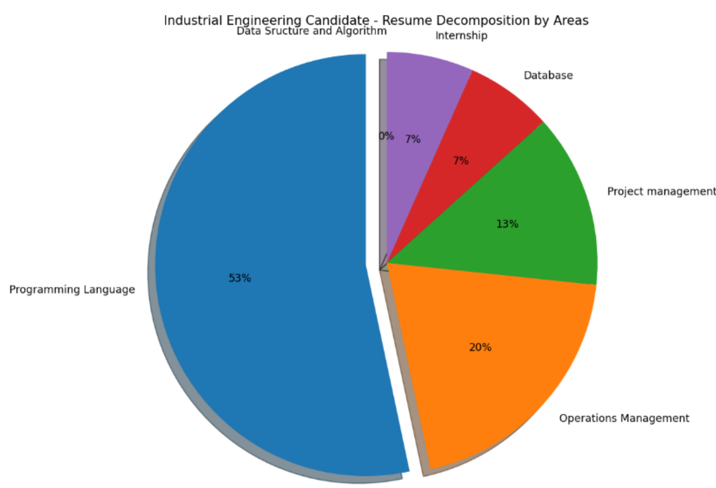


Fig 11. Individual pie-chart analysis

VII. ADVANTAGES AND DISADVANTAGES

Advantages

1. Faster Hiring Process

The system significantly reduces the time required to screen resumes manually. By automating the screening process, recruiters can quickly identify qualified candidates. This leads to a faster and more efficient hiring cycle.

2. Reduces Human Error

Manual resume screening can often lead to mistakes, fatigue, or oversight. This system ensures all resumes are processed without missing any details. As a result, it improves accuracy in candidate selection.

3. Objective Evaluation

The resume screening is based on keyword scoring, which removes subjectivity from the process. It helps maintain fairness by treating every applicant equally. This reduces personal biases that might influence hiring.

4. Improved Candidate Matching

The system uses domain-based scoring to match candidates with the most relevant job profiles. It increases the chances of finding the right fit for a role. This improves the overall quality of hires.

5. Data-Driven Decisions

Admins can view charts and statistics to analyze the performance of candidates and the system. These insights help in making informed recruitment decisions. Visual data adds clarity to the screening results.

6. Handles Bulk Resumes Easily

This system is capable of processing a large number of resumes simultaneously. It's especially useful during campus drives or mass recruitment campaigns. Bulk screening becomes manageable and efficient.

7. Customizable Keywords

Administrators can update and modify the keyword sets as per job roles and requirements. This makes the system flexible and adaptable to changing hiring needs. It allows finetuning for better results.

8. User-Friendly Interface

The interface is simple, clean, and easy to navigate for both recruiters and candidates. Uploading resumes and analyzing results requires minimal training. A good UI enhances the user experience.

9. Secure Resume Storage

All resumes and related data are securely stored in a backend database. Proper security measures prevent unauthorized access. This ensures that user data remains confidential and protected.

10. Cost Saving for Recruiters

By automating screening tasks, the system reduces the need for external recruitment services. It cuts down the time and resources spent on initial shortlisting. This leads to significant cost savings over time.

Disadvantages

1. Limited to PDF Format

The system currently handles only .pdf files for resume uploads.

Candidates uploading .docx or other formats will face compatibility issues.

This limits usability unless format support is extended.

2. No Machine Learning Personalization

The keyword-based scoring does not adapt or learn from feedback.

This static approach may fail to reflect modern resume nuances.

Incorporating ML could improve recommendation accuracy.

3. No Real-Time Admin Notification

Admins do not receive real-time alerts when new resumes are uploaded.

They have to manually check the dashboard for updates.

This delay might affect timely shortlisting in fast-paced hiring.

4. No Resume Download Feature

Currently, the system does not allow admins to download resumes for offline review.

This reduces flexibility for recruiters who prefer reviewing resumes outside the portal.

Adding this feature would improve convenience.

5. Manual Keyword Maintenance

Domain keywords are hard-coded and maintained manually in the system.

This is time-consuming and requires periodic updates.

A dynamic keyword management panel could make this easier.

6. No Feedback Mechanism

There is no option for admins to provide feedback on resume quality or shortlisting criteria.

This makes it difficult to improve the screening logic based on real outcomes.

Integrating feedback could enhance future accuracy.

7. Basic Authentication System

The login system is simple and does not include password encryption or multi-factor authentication.

This poses a risk if the system is deployed in a real-world environment.

Improving security layers is essential for data protection.

8. Performance Limits on Large Uploads

Uploading large PDF resumes may slow down processing or cause delays.

This affects the performance when handling thousands of resumes at once.

Optimizing file handling and preprocessing is necessary for scaling.

9. No Job Role Mapping

Although resumes are analyzed, there is no job posting or job matching module.

The system doesn't recommend roles based on resume content.

This limits its scope to only resume screening and not full recruitment.

10. Lack of Mobile Compatibility

The current frontend may not be optimized for mobile devices.

This reduces accessibility for users trying to access the platform via smartphones.

Responsive design or a mobile app would solve this limitation.

IX. FUTURE SCOPE

The Resume Screening and Job Matching System holds great potential for further development and integration into full-scale recruitment platforms. With the rapid growth of job applications and the need for intelligent hiring tools, this system can evolve beyond basic keyword screening to a fully AI-driven recruitment assistant. Future enhancements can help bridge the gap between job seekers and recruiters more efficiently and accurately.

One of the primary improvements could be the integration of Machine Learning and Natural Language Processing (NLP) techniques to analyze resumes in a more contextaware manner. These models can learn from historical hiring patterns and feedback to rank resumes more intelligently, considering not just keywords but also semantic meaning and experience relevance.

Another potential upgrade is the addition of a job posting and matching module, where candidates can view relevant job openings based on their resume content. This would transform the system from a one-sided screening tool to a two-way recommendation engine for both recruiters and job seekers.

Moreover, implementing automated email notifications, admin alerts, and real-time messaging would streamline the recruitment process, keeping candidates and HR teams informed throughout the hiring stages. Mobile compatibility or a dedicated Android/iOS app can also be developed to allow job application and resume tracking on the go.

Finally, incorporating resume scoring dashboards, candidate ranking systems, interview scheduling, and chatbot assistance could make this system suitable for enterprise-level use in companies, consultancies, and job portals. With cloud integration and data analytics, it could also help in tracking hiring trends and optimizing recruitment strategies over time.

X. CONCLUSION

The Resume Screening and Job Matching System significantly simplifies the recruitment process by automating the initial stage of candidate evaluation. By analyzing uploaded resumes using keyword matching and scoring across multiple job domains, it assists recruiters in identifying the most relevant candidates faster and with greater accuracy. The system also enhances the overall hiring process by reducing manual work and human bias during the screening phase.

The web-based solution developed using Python, Django, MySQL, HTML, CSS, and JavaScript provides a user-friendly interface for both candidates and recruiters. Features like domain-wise resume categorization, PDF parsing, pie chart visualizations, and admin dashboard functions make it a comprehensive platform for preliminary resume analysis and selection. Additionally, the inclusion of dynamic keyword scoring ensures flexibility and adaptability to different job roles.

Overall, this system showcases how intelligent automation can revolutionize traditional hiring methods. While it is currently a prototype, it lays a strong foundation for future enhancements involving deep learning, natural language processing, and real-time job matching. With further development, this solution has the potential to become a powerful tool for efficient, data-driven recruitment across industries.

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